SLIDE 01 OF 07 // TITLE SLIDE

ROCK PAPER SCISSORS

ARCADE E-SPORTS EDITION

A Modern, Interactive Web-Based Multiplayer & AI Arena Game

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Course: **BCA (5th Semester)**

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10 PTS
TOURNAMENT TARGET

MARKOV AI
PATTERN PREDICTION ENGINE

2X CLUTCH
SUDDEN DEATH TIE-BREAKER



SLIDE 02 OF 07 // OVERVIEW

PROJECT OVERVIEW

- **Objective:** To design and develop a modernized, E-Sports arcade version of the classic Rock-Paper-Scissors game using pure frontend web technologies without external frameworks.
- **Tech Stack Used:**
  - → **HTML5 & 2D Canvas:** For structural layout, interactive arena elements, telemetry HUD, and real-time particle explosions.
  - → **CSS3:** For advanced animations, 3D perspective cyber-grid, dynamic theme variables, and responsive media queries.
  - → **Vanilla JavaScript:** For core game logic, state management, Markov AI prediction, Web Audio synthesis, and DOM manipulation.
- **Key Highlights:**
  - → First to 10 points tournament format with **2X Sudden Death** tie-breaker rounds.
  - → Cross-platform playability (supports both Touch UI and Arcade Keyboard controls).

LAYER	TECHNOLOGY	ROLE IN ARCADE E-SPORTS EDITION
Structure & FX	HTML5 + `<canvas>`	Lobby modal, Jumbotron scoreboard, action pads, and 60 FPS neon particle bursts.
Presentation	CSS3 Custom Variables	3D rotating grid floor, CRT scanlines, and instant 3-theme Cyberpunk palette switching.
Game Engine	Vanilla ES6+ JavaScript	Markov Chain AI, Web Audio API oscillator SFX, 5s shot clock, and `localStorage` stats.

SLIDE 03 OF 07 // GAME MODES & AI

GAME MODES & CONTROLS

- **Versatile Gameplay Modes:**
  - → **VS BOT Mode (Dual AI Difficulty):** Features both a standard **RNG Bot** and a **Markov Chain Pro AI** that tracks Player 1's transition matrix to predict and counter repetitive move patterns.
  - → **PVP Mode (Player vs Player):** Local multiplayer allowing two players to compete head-to-head on the same device.
- **Dynamic Control Schemes:**
  - → **Touch Mode:** Interactive on-screen neon buttons optimized for mobile screens and mouse clicks.
  - → **Keys Mode:** True arcade keyboard integration (Player 1 uses `A, S, D`; Player 2 uses `J, K, L`).
- **5-Second Shot Clock Timer:** Optional tournament countdown clock (⏱️ `CLOCK: ON`) that forces fast decision-making and auto-plays a move if time expires.



SLIDE 04 OF 07 // UI/UX DESIGN

UI/UX & VISUAL DESIGN (CSS3)

- **3D Cyber-Grid Background & CRT Scanlines:** Features an animated 3D perspective grid floor (`rotateX(75deg)`), radial dark-space gradients, and a subtle CRT scanline overlay for an authentic retro-future arena vibe.
- **Live Cyberpunk Theme Switcher:** Uses CSS custom properties (`--p1` , `--p2`) to switch instantaneously between **Cyber Blue**, **Toxic Matrix Green**, and **Synthwave Gold** neon color schemes.
- **Responsive Tabletop Mobile Design:** Uses CSS Media Queries (`@media max-width: 768px`) to convert the arena into a vertical layout and rotates Player 1's controls `180deg` for face-to-face mobile play.
- **Interactive Jumbotron & Telemetry Bar:** Displays health-style progress bars, recent clash history chips, real-time Win Rate %, and move usage distribution.

THEME 01	THEME 02	THEME 03
CYBER BLUE (#00D2FF / #FF0055)	MATRIX GREEN (#00FF88 / #BD00FF)	SYNTHWAVE GOLD (#FFCC00 / #00E5FF)

SLIDE 05 OF 07 // CORE LOGIC

CORE GAME LOGIC (JAVASCRIPT)

- **State Management & LocalStorage Persistence:** Tracks active game modes, scores, streaks, and move distribution dynamically while saving all-time Career Wins and Best Streaks in browser `localStorage`.
- **Action Locking (Anti-Cheat):** Once a player selects a move, their action pad hides and displays a `LOCKED //` badge, preventing input spamming and screen-peeking in PVP mode.
- **Turn Evaluation & 2X Sudden Death:**
  - → Compares choices using conditional logic after both players lock in, and triggers a **2X Double Points** multiplier whenever a round ends in a Tie.
  - → Unlocks live achievement toast notifications for major in-game milestones.

ACHIEVEMENT BADGE	TRIGGER CONDITION	VISUAL FEEDBACK
🥇 FIRST BLOOD	Winning the first point of the match	Slide-in neon toast notification
⚡ CLUTCH BREAKER (2X)	Winning a 2X Sudden Death round after a Tie	Awards +2 PTS + Golden Toast
🔥 UNSTOPPABLE	Achieving a 3-round consecutive win streak	Streak flame badge + Toast
🧠 MARKOV AI SLAYER	Defeating the Predictive Pro AI (10 PTS)	Mastery achievement + Career save

ANIMATIONS & POLISH

- **Web Audio API Synthesizer (Zero External Assets):** Generates real-time retro arcade sound effects (clicks, clash win/loss tones, tie buzzers, and victory fanfare) using pure JavaScript `AudioContext` oscillators.
- **HTML5 Canvas Particle Engine & Screen Shake:** Spawns 32 glowing neon particles at the center of the arena on every clash using `requestAnimationFrame` alongside a CSS keyframe screen-shake effect.
- **Cubic-Bezier Clash Animations & Score Popping:** Uses custom `cubic-bezier(0.175, 0.885, 0.32, 1.275)` timing functions for emoji clashes and dynamic scale-up score transitions.
- **Bot Delay Simulation:** In VS BOT mode, a `400ms` delay with an hourglass emoji (⏳) simulates the AI "thinking" before revealing its counter-move.

SOUND EVENT	OSCILLATOR WAVEFORM	FREQUENCY & DURATION PROFILE
Move Lock / UI Click	`sine` wave	520 Hz crisp short pulse (80ms)
Round Victory	`triangle` wave	Dual-tone ascending chime (440 Hz → 660 Hz)
Round Defeat / Tie	`sawtooth` / `square`	220 Hz low buzz (defeat) / 320 Hz alert (tie)
Tournament Victory	`triangle` arpeggio	4-note arcade fanfare (440, 554, 659, 880 Hz)

SLIDE 07 OF 07 // CONCLUSION & FUTURE SCOPE

CONCLUSION & FUTURE SCOPE

- **Conclusion:** Successfully created an engaging, fast-paced, and visually appealing E-Sports web game combining predictive Markov AI, synthesized Web Audio, Canvas particle physics, and persistent telemetry using only fundamental web technologies (HTML5, CSS3, and Vanilla JS).
- **Future Enhancements:**
  - → Implementing WebSockets (`Socket.io`) for real-time online multiplayer across different devices.
  - → Adding a global cloud leaderboard and ELO tournament bracket system.
  - → Packaging the application as an installable Progressive Web App (PWA) for offline mobile play.

THANK YOU!

ANY QUESTIONS? (Q&A)

Live Demo: https://dipanshurana121.github.io/rock-paper-scissors-game/

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